

Bastien Gauthier-Soumis

Computing tinkerer

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Summary

- Multidisciplinary interest rooted in personal and professional experience in applied physics, computer science, technological arts and digital medias
- Research experience in the field of computational physics and modeling
- Technical (software and hardware) design experience within research-creation projects
- Sociability and appreciation for collective work developed within a number of artistic collaborations and academic and community organisations
- Experience working and communicating complex scientific and technical ideas in French (native French-Canadian speaker) and English (fluent speaker)

Formation

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| • Diplôme d'études supérieures spécialisées en Arts, Création et Technologies
(Specialized graduate studies in Arts, Creation and Technologies)
Université de Montréal, Montréal, Québec, Canada
30 credits, 4.26/4.3 average | 2020-2021 |
| • Baccalauréat en Génie Physique
(Bachelor in Engineering Physics)
École polytechnique de Montréal, Montréal, Québec, Canada
120 credits, 3.73/4.0 average | 2016-2020 |
| • Exchange studies in Engineering Physics
Aalto University, Espoo, Finlande | 2018-2019 |
| • Diplôme d'études collégiales en sciences de la nature
CÉGEP de St-Hyacinthe, St-Hyacinthe, Québec, Canada | 2014-2016 |

Expertise and skills

- Experience applying knowledge of physics and technical principles ranging from classical mechanics, quantum mechanics, quantum field theory, electromagnetism, statistical physics, quantum informatics, analysis, linear algebra and numerical computing methods within numerous academic and creative contexts
- Experience in choosing and developing algorithms, data representation and software with various programming languages (*Python3.5*, *Fortran77/90/2008*, *C*, *C++*, *Java*, *Lua*, *SHELL*)
- Experience in autonomous artistic and scientific research and collaborating through continuous integration and iterative contribution
- Experience in using digital infrastructures for sharing data and organising work (from scientific computing on shared supercomputers to hosting a research-creation collaborative web platform and agile development through *Git* version control)
- Experience in designing, troubleshooting, maintaining and repairing autonomous and resilient digital systems and technological devices for multiple artistic installs and research-creation projects, with an emphasis on developing and expressing their creative and conceptual elements
- Familiarity with various digital arts mediums, in particular the use of microelectronics and physical computing devices in artistic creation contexts and interactive art installs

Past projects and positions in the field of computational physics

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|---|------|
| • Research project, internship
Study of spin waves in magnonic crystals using experimental measurement and computing models within the magnetism laboratory of Polytechnique Montréal | 2018 |
| • Research project, internship
Development of a computational model to investigate the extinction spectrum of nanosponges within the LP ² L research group of Polytechnique Montréal | 2017 |

Past projects and positions in the field of technological arts

- **Technical director for a research-creation collective** 2021-2025
Being in charge of technical aspects of all projects developed by the *Critical Gardening Collective*, a research-creation group joining artists and researchers from University of Montréal Communication Department in investigating the experience of the world of mosses as a privileged viewpoint to question life within the antropocene.
 - Design of technological systems integrated to digital art installations adapted to the harsh climate of the tundra, including a wind power system, weather sensors specifically tuned to the bryophytes environment and a LoRa mesh network
 - Fabrication and integration of those hardware and software systems within an autonomous digital device
 - Assistance to the installation of the artistic installs during residencies at the University of Helsinki Biology research station at Kilpisjärvi (2023) and at the Cape Horn Internacional Center in Puerto Williams (2025) and in preparation to those research group residencies; in particular, assembly of electronics circuits, integration of the technological devices within outdoor sculptural elements and on-site troubleshooting and tuning of the devices
 - Responsibility of the long term support of the technological devices within the installs and supervision of their maintenance
- **Technical expertise for the design of an artistic installation** 2024
Assistance for the technical design and manufacturing of electromechanical elements included in artworks that were part of the exhibit *Thing(s)* by artist Neïtah Janzing at Schaubude Berlin
 - Design and manufacturing of motorized mechanism to integrate movement to a sculpture
 - Design and programming of a generative digital control system to animate the sculpture according to multiple parameters to be set by the artist via a web interface
- **Technical expertise in a research residency in contemporary dance** 2023
Assistance for the design and manufacturing of technological art elements to be integrated to a contemporary dance performance by artist Fjord Gagnebin within the context of their residency at Studio Parbleux in Montréal
 - Conception and preparation of mechanical scenographic elements including digital control
 - Set up of the scenographic elements and real time control of them in coordination with the dance performance during its presentation
- **Direction for the adaptation and set up of an artistic installation** 2023
Installation of the multidisciplinary work *Monstra Te Esse Matrem* by bioartist François-Joseph Lapointe within a new exhibition context
 - Preparation and adaptation of the art installation *Monstra Te Esse Matrem* for its exhibition at the Université de Montréal exhibition center according to the initial direction of the artist that designed it
 - Coordination of the set up, storage and transportation of the work after its exhibition
- **Technical direction for some interactive elements of a multidisciplinary artwork** 2022
Technical expertise for the integration of technological art elements within the multidisciplinary art installation *Monstra Te Esse Matrem* by bioartist François-Joseph Lapointe
 - Design of technological devices integrated to an art installation (interaction, digital control and automation of an hydraulics system)
 - Preparation, adaptation and installation of the *Monstra Te Esse Matrem* art installation for its first exhibition at the *Bang! Art center*
- **Technical direction for an outdoor digital arts exhibition** 2021
Technical direction of the traveling exhibition *Ordinaire de Nature* that was installed in four parcs of the city of Laval in collaboration with artist-researcher Oria Steenkiste and design studio Boume
 - Design and manufacturing of rugged electromechanical devices to integrate in dynamic, networked and interactive art installations reacting to their outdoor environment according to the artistic director instructions
- **Connexion{s}, exposition d'art technologique** 2021
Participation to the exhibition of the 2021 graduate of the D.E.S.S. in arts, création et technologie taking place in the science library of the Université de Montréal
 - Completion of a research-creation project investigating the materiality of digital machines, infrastructures and processes according to a critical and participative approach in the line of *critical making*
 - Design and manufacturing of a digital installation where custom macroscopic symbolic inscription mechanisms acquire, process, transcribe and store digital data about their environment
 - Participation to the collective design of a technological art exhibition and of a publication about it

- **Photograph exhibit** 2018
Participation to the collective exhibition *Microscopic view: Experimental Light Images* presented within the course *UWAS-C0037 Microscopic view* at Aalto University
- Realization of a visual artwork using scientific imagery tools to experiment with their technical particularities to investigate and contrast the different contexts of image production and interpretation, as well as the diversity of functions that images have in different fields

Grants and prizes

- **Undergraduate research grant** 2017,2018
Natural Sciences and Engineering Research Council of Canada
-Academic excellence
-Support from a professeur holding a NSERC research grant

Languages

-French, native speaker -English, fluent